Understanding Sleep Factors and The Impact

SCM 315

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Sources Used For This Project

Source: https://www.kaggle.com/datasets/adilshamim8/sleep-cycle-and-productivity

Information Gained:

- - Total Sleep Hours
- -Sleep Quality
- -Exercise (min/day)
- -Caffeine Intake (mg)
- -Work hours (hrs/day_
- -Productivity Score
- -Mood Score

Sleep Quality Insights

Sleep Quality Insights

Visualization :

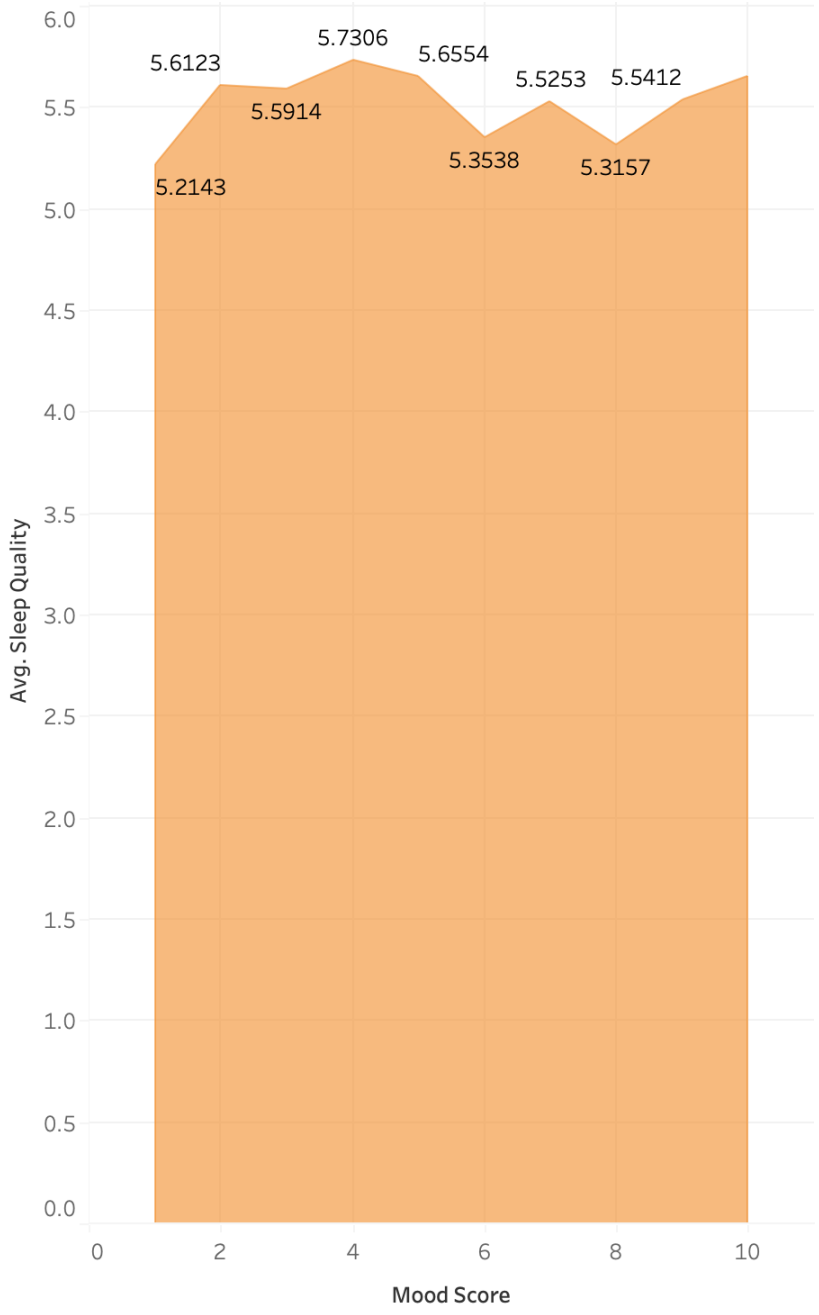
- This data set is comprised of average sleep quality against mood scores and exercise
- The X-axis on the Area graph (graph on the left) represents the mood score of participants, ranging from 0-10
- The Y-axis on the Area graph represents the average sleep quality of participants, ranging from 0-10

- The X-axis on the dual-axis combination graph (graph on the right) represents exercise min/day
- The Y-axis is a dual axis with the avg sleep quality of participants on the right side ranging from 0-6.5 , and the avg mood score of participants on the left side, ranging from 0-7

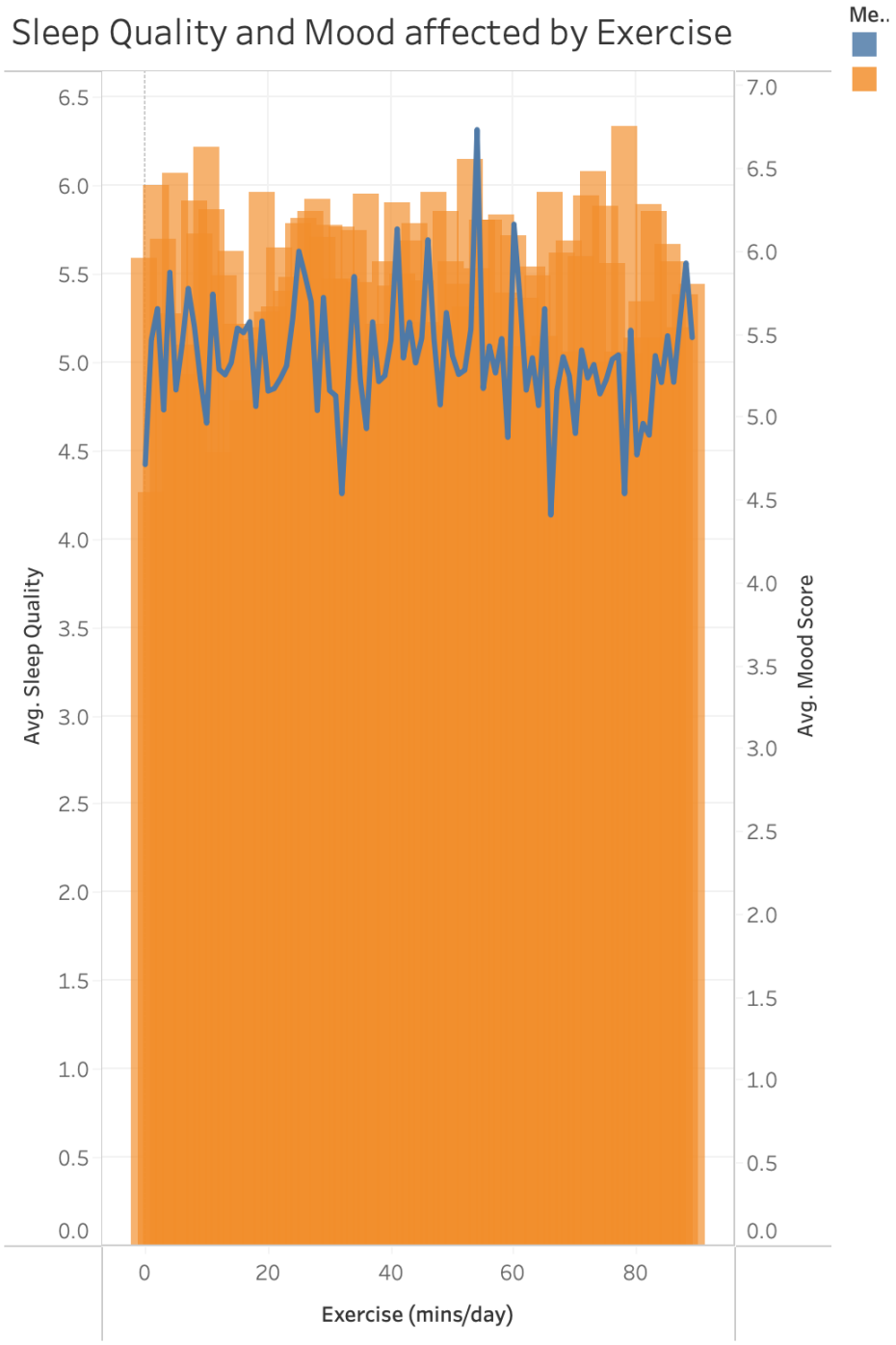
Reasoning:

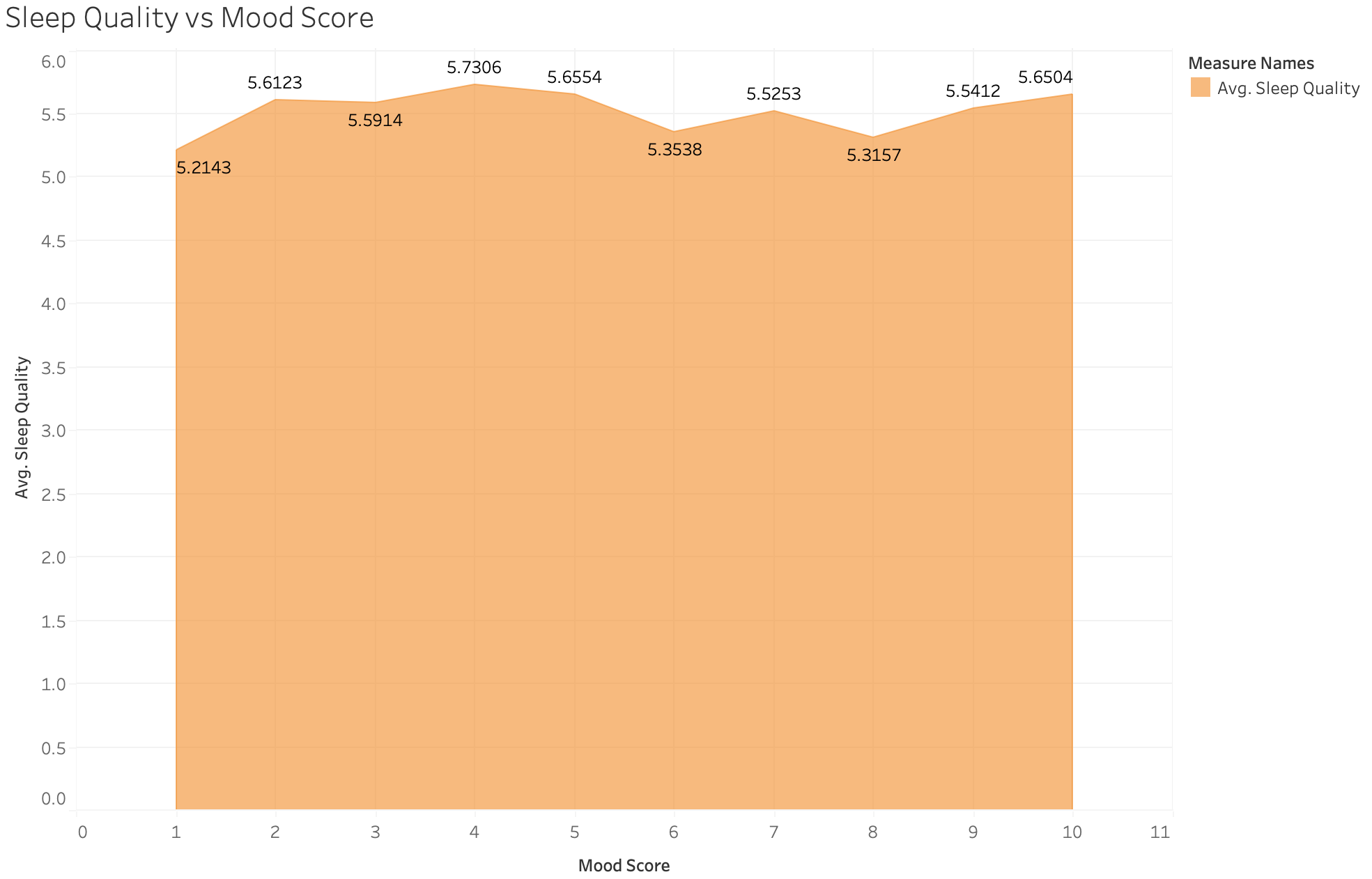
- This visualization was created to show how various factors such as exercise, mood, and sleep quality impact each other, highlighting the correlation between sleep quality and mood in relation to exercise levels.
- The dual axis combination graph makes it easier to see the fluctuation in sleep quality and mood across different amounts of exercise, helping to identify trends in how exercise influences both.
- The area graph helps to effectively compare the two metrics, sleep quality and mood, while also showing how they are affected by exercise, offering a comprehensive view of these interconnected factors.

Sleep Quality vs Mood Score



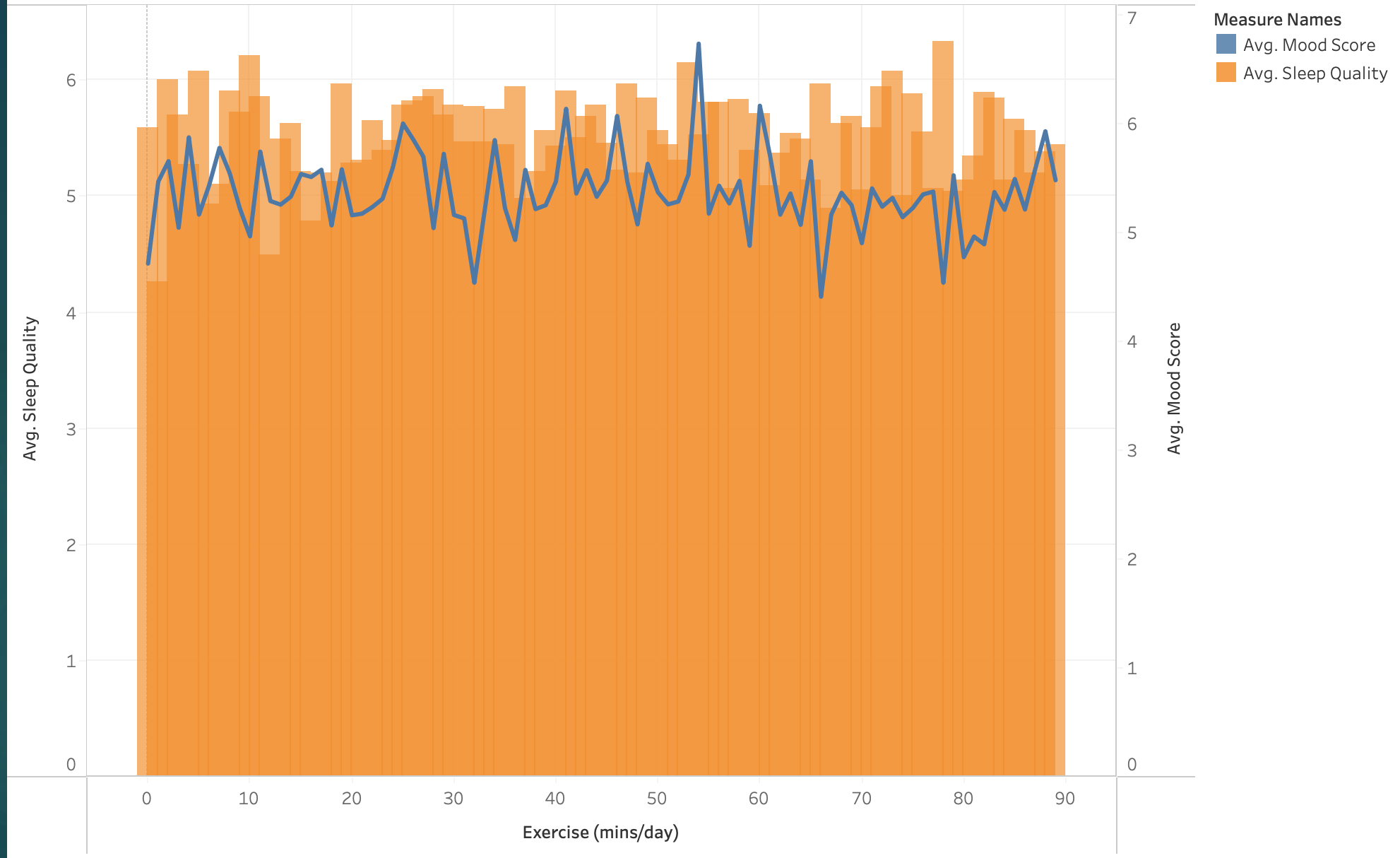
Sleep Quality and Mood affected by Exercise





The plot of Avg. Sleep Quality for Mood Score. Color shows details about Avg. Sleep Quality.

Sleep Quality and Mood affected by Exercise



The trends of Avg. Sleep Quality and Avg. Mood Score for Exercise (mins/day). Color shows details about Avg. Sleep Quality and Avg. Mood Score.

Conclusion

Sleep Quality vs. Mood Score:

- The average sleep quality fluctuates across different mood scores. The highest sleep quality is consistently observed when mood scores are between 5 and 6, with sleep quality gradually decreasing as mood scores approach 10.
- **Key Insight:** Sleep quality seems to improve with higher mood scores, suggesting that better sleep leads to a positive emotional state. This relationship indicates the importance restful sleep for one's mental well-being.

Sleep Quality and Mood Affected by Exercise:

- Sleep quality and mood are both positively influenced by exercise, but with varying intensities. Moderate exercise (20-40 minutes) correlates with the highest sleep quality, while intense exercise beyond 60 minutes seems to reduce sleep quality slightly.
- **Key Insight:** Moderate exercise appears to be the sweet spot for improving both mood and sleep quality, while excessive exercise might cause restlessness or fatigue, affecting sleep negatively. This pattern suggests that finding a balance in physical activity can optimize sleep and mood.

Conclusion on Sleep Quality Insights:

- Better sleep tends to result higher mood scores , highlighting the significant connection between emotional rest and well-being.
- Exercise can improve sleep quality and mood, but moderation is crucial—too little exercise may not have a significant impact, while too much exercise can be counterproductive.
- These insights emphasize the need for balanced lifestyle choices, particularly focusing on emotional health and moderate physical activity, to achieve optimal sleep quality.

Work and Sleep Patterns

Visualization :

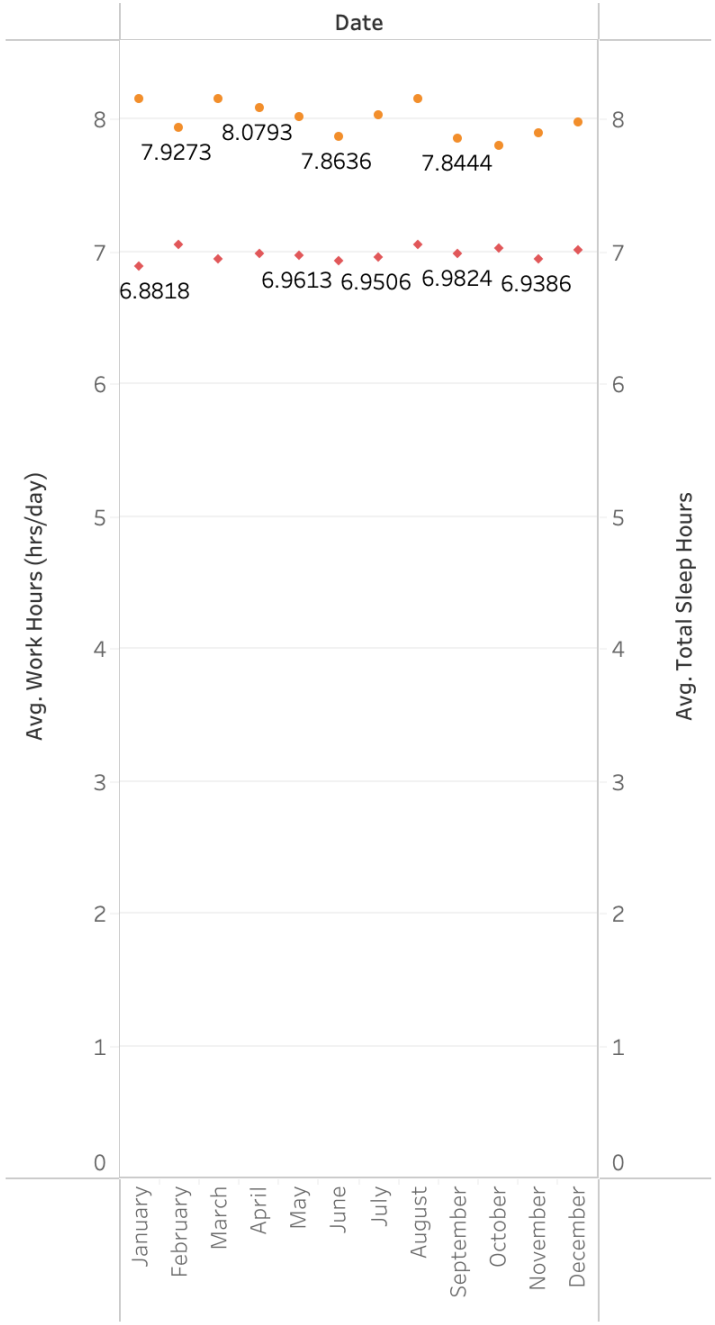
- This data set is comprised of participants caffeine intake, average work and sleep hours, and age
- The first graph is a line graph with scatter plot markers. The X-axis is a dual axis and is comprised of participants average work and sleep hours. The axis ranges from 0-8
- The Y-axis represents the months, ranging from Jan-Dec

- The second graph is a stacked bar graph. The X-axis represents the days of the week. Ranging from Sunday-Saturday.
- The Y-axis represents total sleep quality data. Ranging from 0-4000
- Each bar represents a day of the week, and the segments within the bars are the values for each week. The different shades of blue represent varying levels of sleep quality.

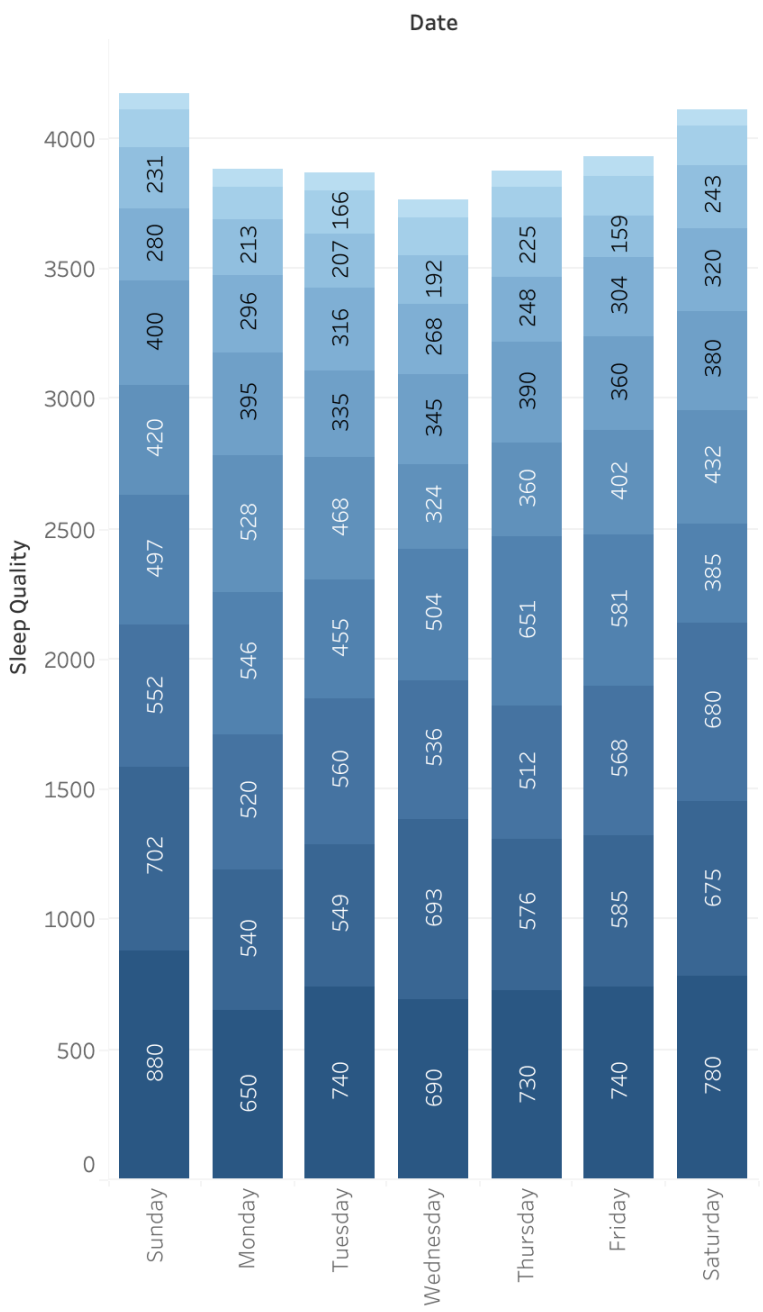
Reasoning:

- This dashboard was created to examine how work hours and weekday patterns affect overall sleep quality. The combination of the scatter plot and the stacked bar chart provides a visual narrative of the interplay between work and sleep. The scatter plot helps illustrate the negative correlation between the number of work hours and sleep hours, while the weekday sleep quality chart showcases how work stress or routine might affect rest patterns throughout the week.
- These visualizations offer a holistic understanding of the work-sleep relationship, highlighting how work hours can impact sleep quality and how individuals may experience better sleep quality during weekends when work hours are typically lower.

Avg Work Hrs vs Avg Total Sleep Hours



Weekday vs Sleep Quality



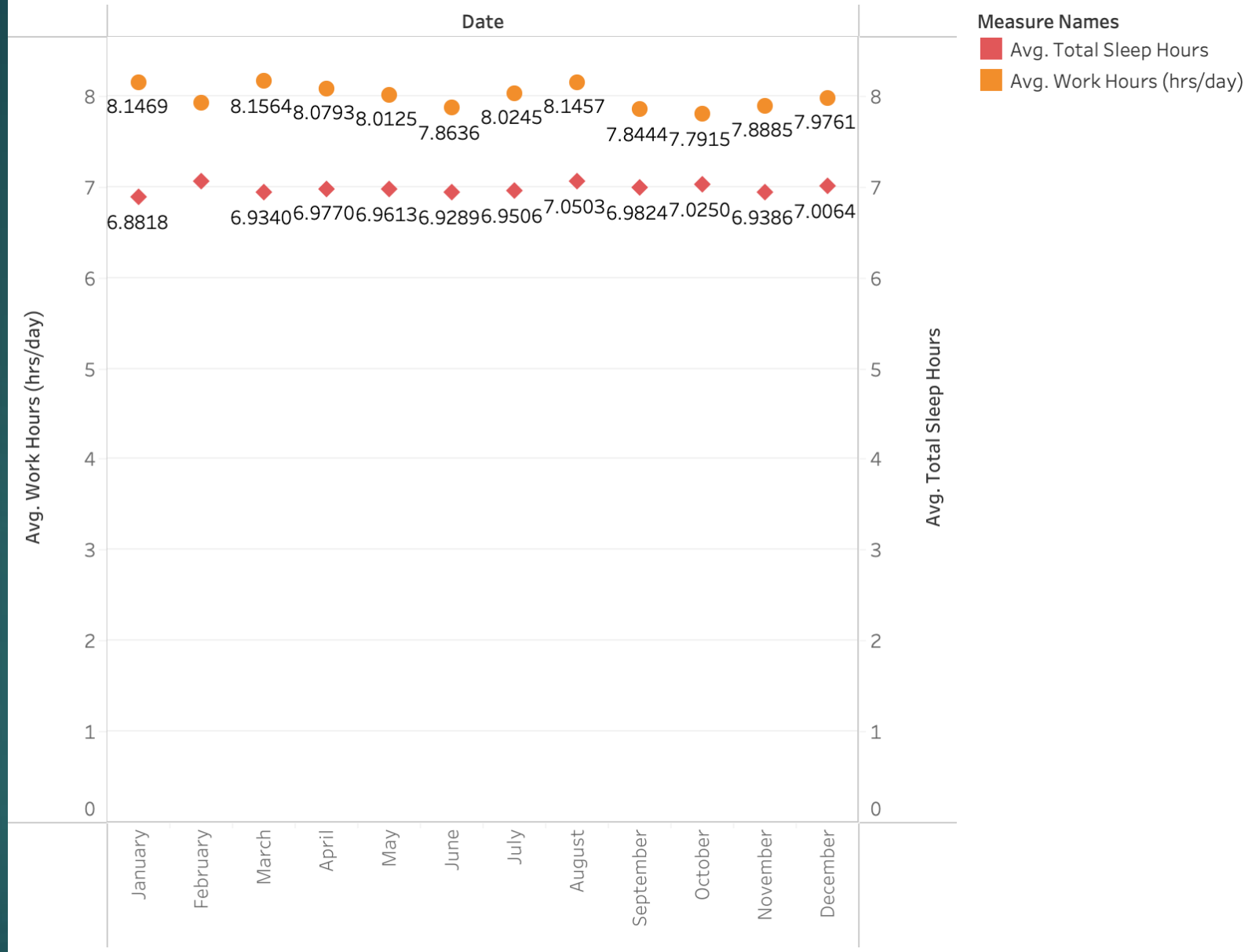
Measure Names

- Avg. Total Sleep Ho..
- Avg. Work Hours (h..

Sleep Quality

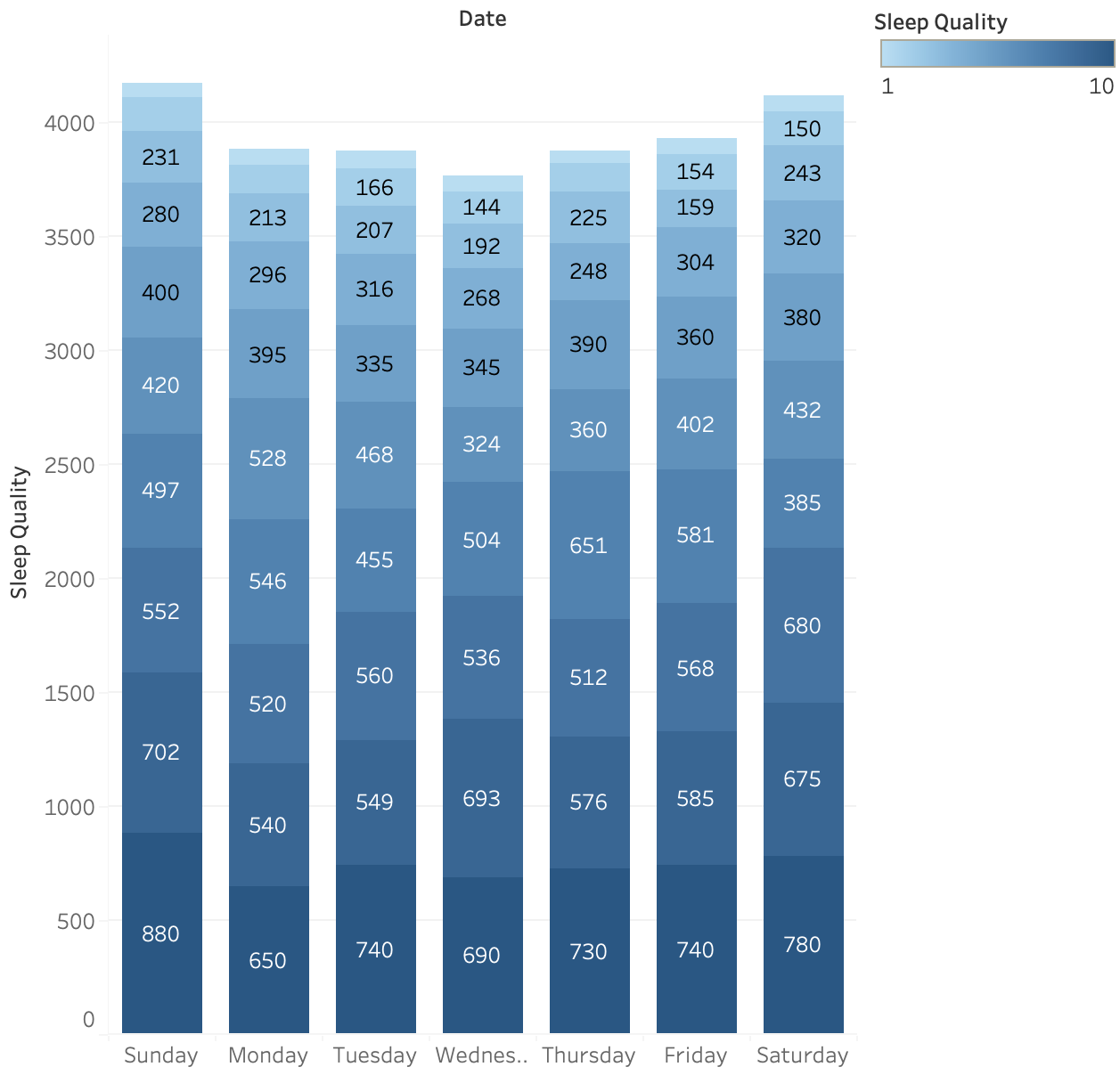
1 — 10

Avg Work Hrs vs Avg Total Sleep Hours



Avg. Work Hours (hrs/day) and Avg. Total Sleep Hours for each Date Month. Color shows details about Avg. Work Hours (hrs/day) and Avg. Total Sleep Hours. For pane Average of Work Hours (hrs/day): The marks are labeled by Avg. Work Hours (hrs/day). For pane Average of Total Sleep Hours: The marks are labeled by Avg. Total Sleep Hours.

Weekday vs Sleep Quality



Sum of Sleep Quality for each Date Weekday. Color shows details about Sleep Quality.
The marks are labeled by sum of Sleep Quality.

Conclusion

Work Hours vs. Total Sleep Hours:

- The data shows a clear trend where higher average work hours correlate with lower total sleep hours. Specifically, months with more work/familial obligation hours, such as those around the holiday season, consistently report reduced sleep by about 1.5 hours on average.
- **Key Insight**: Longer work hours can lead to shorter sleep durations, which may affect overall health and well-being.

Weekday vs. Sleep Quality:

- Sleep quality is consistently lower on weekdays, with a noticeable dip around the midweek (Wednesday and Thursday). On weekends, sleep quality improves by about 0.5 points, which suggests that participants have more time for sleep on the weekends and want to get more sleep in when they can.
- **Key Insight**: Weekdays tend to bring more stress and less sleep quality, while weekends allow for recovery, leading to improved sleep and restfulness.

Impact of High Work Hours on Sleep Quality:

- The analysis also shows that in months with increased work and obligation hours, such as December, show a drop of 0.8 points in sleep quality.
- **Key Insight**: Prolonged work hours significantly impact sleep quality, indicating the importance of achieving a balance between work and rest for mental and physical health.

In short:

- Longer work hours are linked with reduced sleep quality and total sleep hours, demonstrating the importance of balancing work demands with sufficient rest.
- Weekends provide an opportunity for recovery, as sleep quality significantly improves, which highlights the role of rest in maintaining a healthy work-life balance.
- These insights emphasize the need for individuals to manage their work hours to prevent the negative impact on sleep, particularly in high-demand periods, to achieve optimal mental and physical health.

Sleep and Caffeine

Visualization :

- This data set is comprised of caffeine intake, average sleep quality, participant productivity scores, and age
- The first graph is a dual axis line graph. The X-axis represents age of participants. It ranges from 15-60
- The Y-axis is a dual axis, with the left side being caffeine intake, ranging from 0-150. The right side represents the average sleep quality of participants, ranging from 0-6.

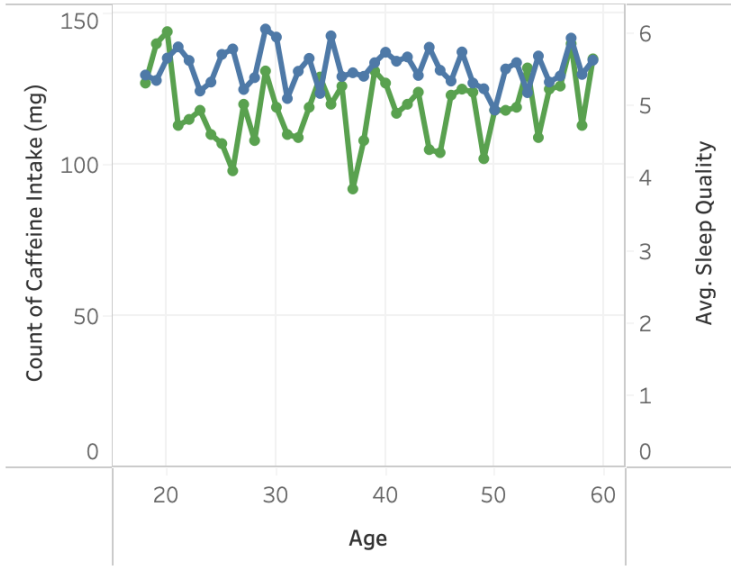
- The second graph is a stacked bar chart that displays average sleep quality and average work hours, with caffeine intake in mg as a separate variable. The X-axis shows the days of the week
- The Y-axis tracks both sleep quality/work hours (ranging from 0-14) and caffeine intake (ranging from 0-140).

- The third graph to the right is a scatter plot. The X-axis represents the participants self described productivity scores. This ranges from 0-10.
- The Y-axis represents participants caffeine intake, this ranges from 0-600.

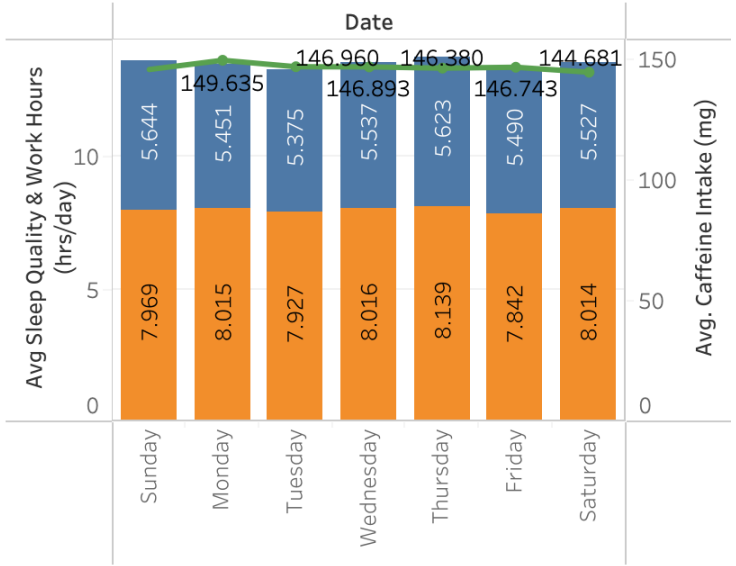
Reasoning:

- This visualization was created to demonstrate how various factors like productivity, caffeine intake, and sleep quality interact with each other, emphasizing the relationship between productivity and caffeine intake across different age groups.
- The line graph was used to track the fluctuations in both caffeine intake and sleep quality by age. This helps to clearly display how these two variables move together or independently across age groups, giving a clearer understanding of how caffeine intake may affect sleep quality in different age categories.
- The stacked bar graph was introduced to show how the interaction between caffeine intake, work hours, and sleep quality varies across the days of the week. This allows us to better understand how these factors fluctuate and correlate with one another on a daily basis, providing insights into daily patterns
- The scatter plot was chosen to show the relationship between caffeine intake and productivity scores because it effectively visualizes how changes in caffeine consumption correlate with shifts in productivity levels. It helps identify any potential trends or patterns across different age groups, allowing us to better understand the impact of caffeine on productivity.

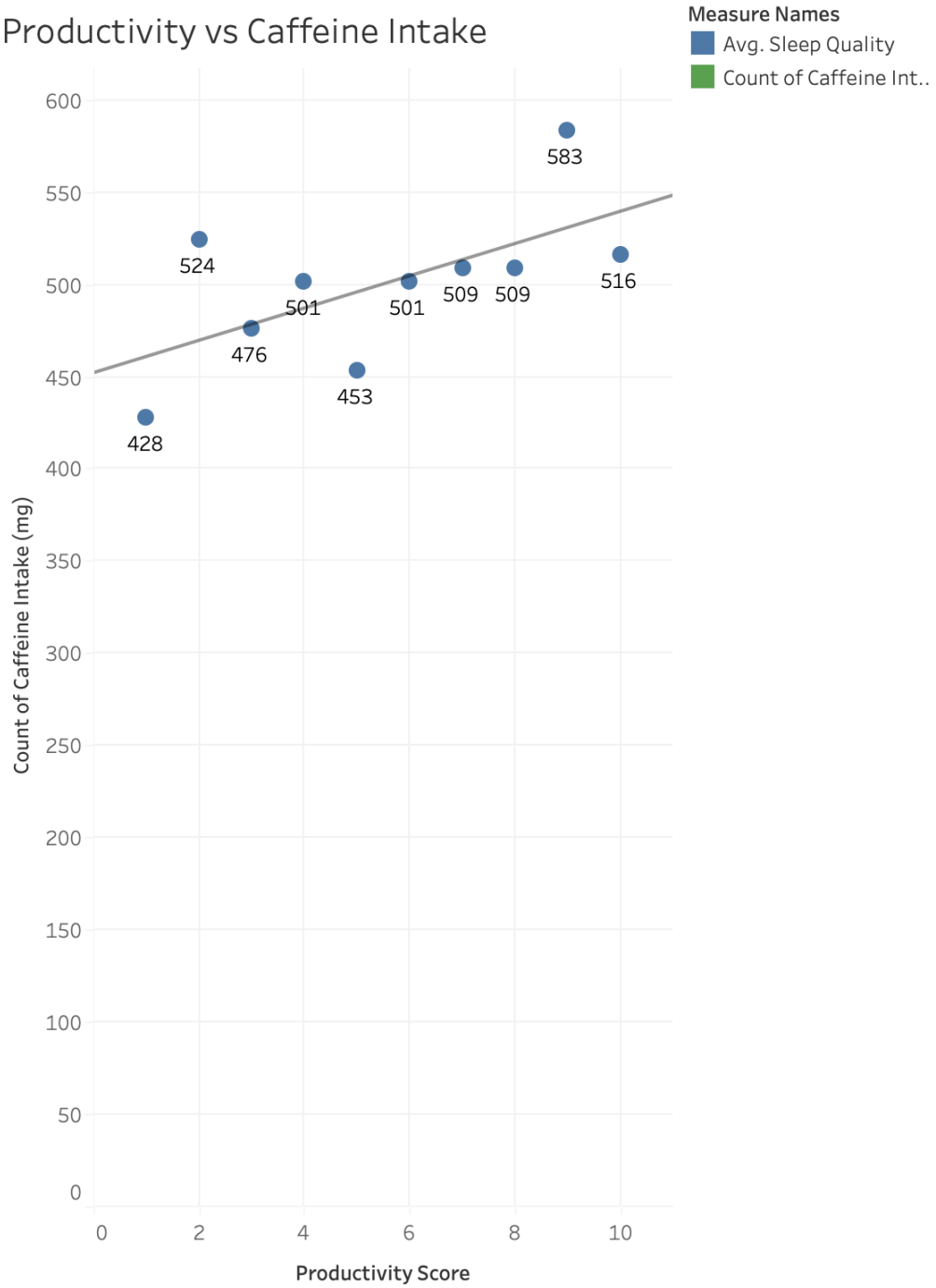
Avg Sleep Quality vs Caffiene Intake- By Age



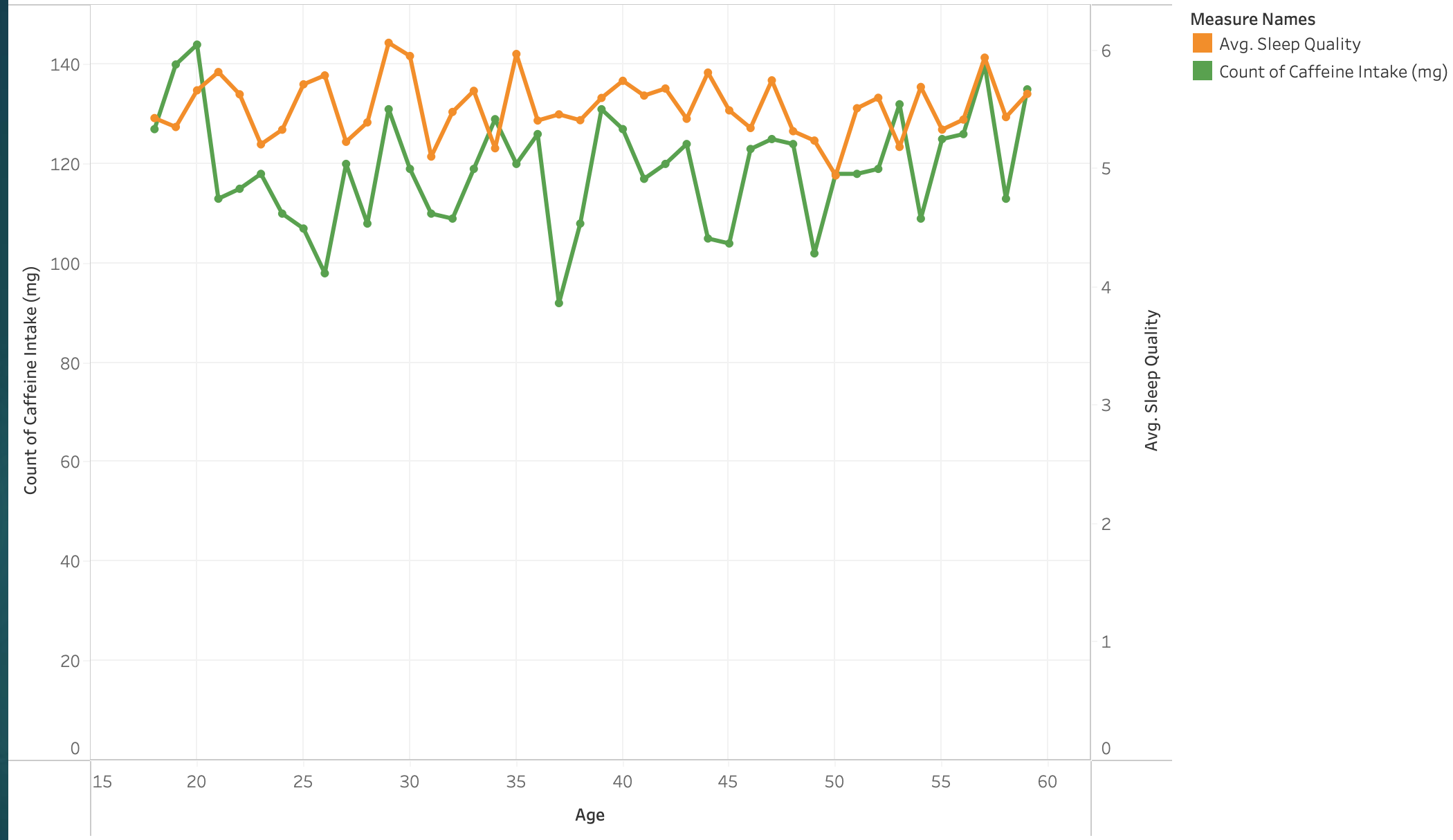
Avg Sleep Quality & Work Hours (hrs/day) vs Avg Caffeine Intake



Productivity vs Caffeine Intake

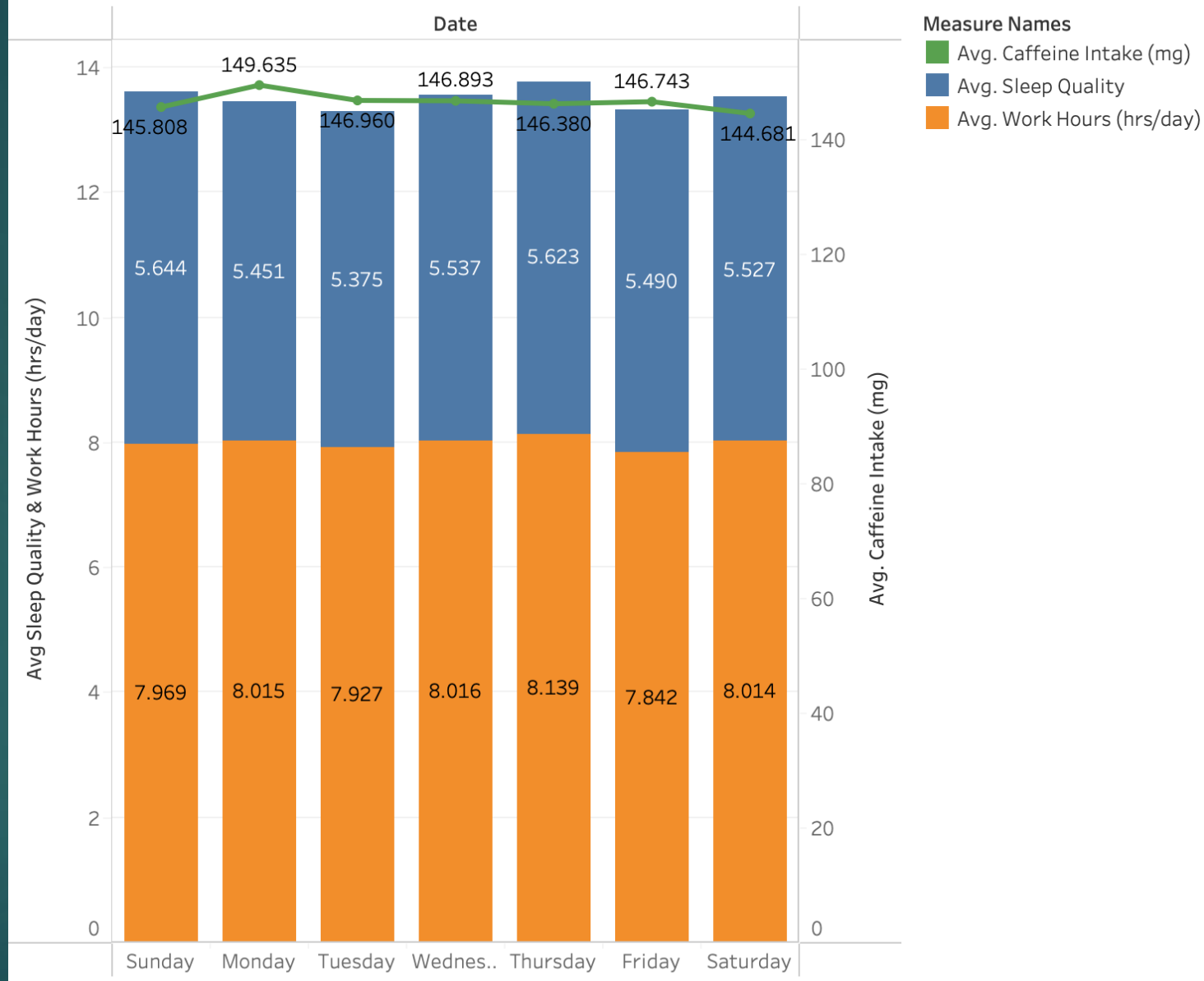


Avg Sleep Quality vs Caffiene Intake- By Age

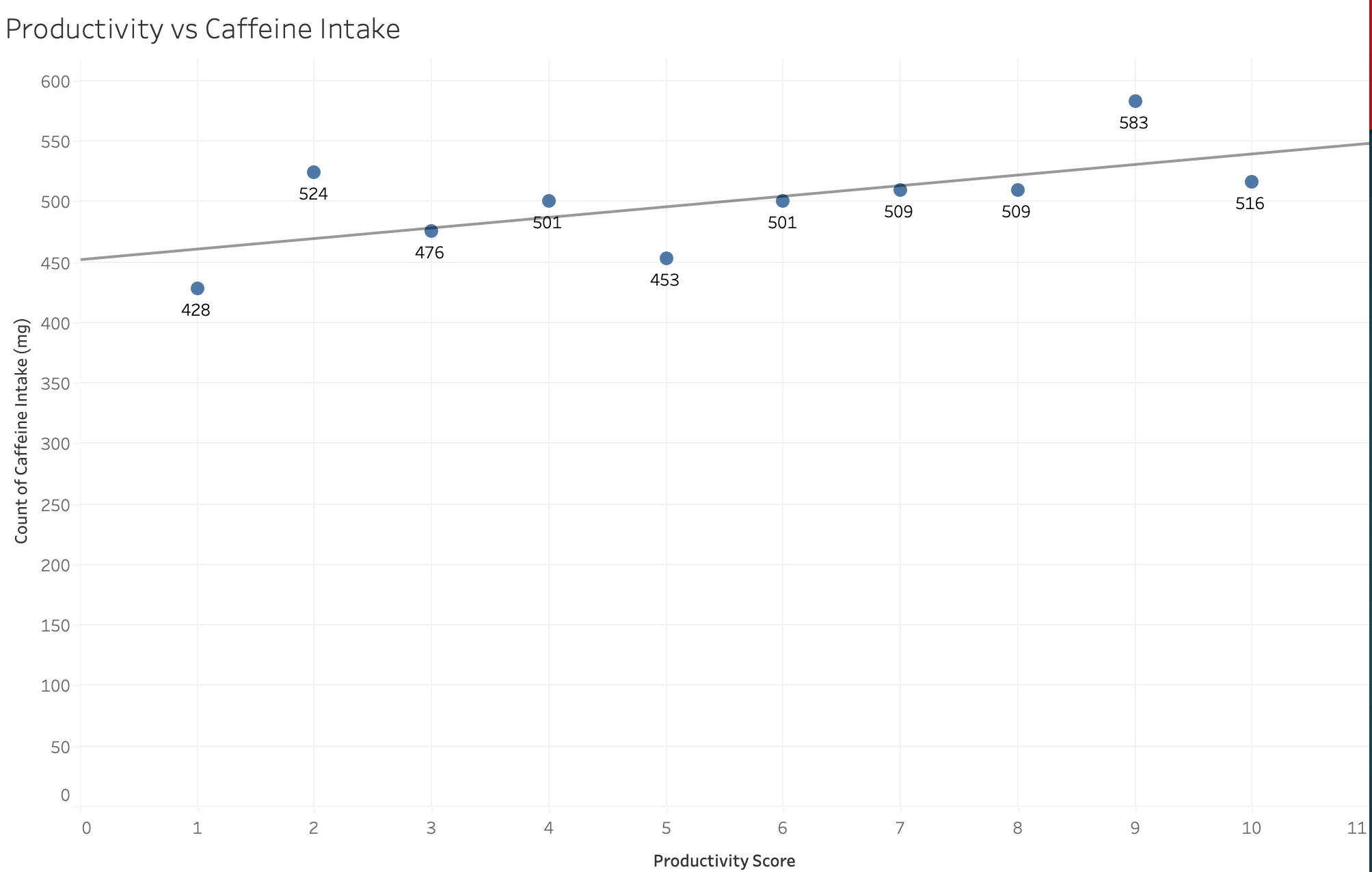


The trends of count of Caffeine Intake (mg) and Avg. Sleep Quality for Age. Color shows details about count of Caffeine Intake (mg) and Avg. Sleep Quality. The data is filtered on Gender, which keeps Female, Male and Other.

Avg Sleep Quality & Work Hours (hrs/day) vs Avg Caffeine Intake



The trends of Avg. Sleep Quality, Avg. Work Hours (hrs/day), Avg. Caffeine Intake (mg) and Avg. Caffeine Intake (mg) for Date Weekday. Color shows details about Avg. Sleep Quality, Avg. Work Hours (hrs/day) and Avg. Caffeine Intake (mg).



The plot of count of Caffeine Intake (mg) for Productivity Score.

Conclusion

Avg Sleep Quality vs. Caffeine Intake by Age:

- The data shows a clear relationship between caffeine intake and sleep quality. Higher caffeine consumption generally corresponds with slightly lower sleep quality across various age groups. Younger individuals (ages 15-30) tend to consume more caffeine, and this trend is reflected in the fluctuating sleep quality data.

- **Key Insight:** Increased caffeine intake may correlate with slightly lower sleep quality, especially in younger age groups, suggesting that while caffeine may help improve alertness, it could be negatively affecting sleep.

Avg Sleep Quality & Work Hours vs. Caffeine Intake (Stacked Bar Chart):

- The chart reveals that caffeine intake appears to have a consistent level throughout the week, with a slight peak in consumption toward the middle of the week. This indicates that people tend to consume more caffeine on weekdays, possibly to combat fatigue and improve focus during work hours.

- On the other hand, sleep quality seems to fluctuate more significantly across the week. It tends to decrease slightly on weekdays (especially toward the middle of the week), possibly reflecting the increased caffeine consumption, which may affect sleep patterns.

- The work hours trend seems to align with caffeine intake, showing a peak during the workweek (Monday through Friday), with a drop on weekends. This suggests that individuals tend to work more during the week, likely requiring more caffeine to maintain productivity and work performance

- **Key Insight**: The stacked bar chart provides a clear visualization of how caffeine intake interacts with both average sleep quality and work hours throughout the week. The chart highlights the variation in these metrics across different days, allowing us to observe if there are specific days when caffeine intake is higher and whether it impacts sleep quality or work hours.

Productivity vs. Caffeine Intake:

- The analysis of productivity scores against caffeine intake reveals a positive trend, with higher caffeine consumption linked to increased productivity. As productivity scores rise, caffeine intake also increases, indicating that people with higher productivity levels tend to consume more caffeine.

- **Key Insight:** Moderate caffeine intake seems to be associated with higher productivity, suggesting that caffeine can be an effective stimulant for boosting work performance.

In Short:

- Increased caffeine intake is associated with a slight reduction in sleep quality, particularly in younger individuals, highlighting the potential trade-off and risks between caffeine use and sleep quality.

- Higher caffeine consumption is also positively linked with productivity, suggesting that moderate caffeine intake can enhance work performance.

- These insights emphasize the need to find a balance in caffeine consumption to optimize both productivity and sleep quality, especially for younger age groups, who need to manage their caffeine intake carefully to avoid sleep disruptions and support overall health.